

RISC-V Instruction Formats

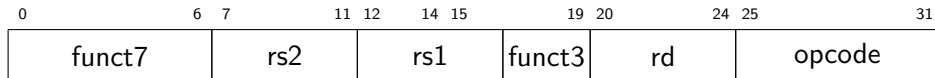
RV32I R-Type vs. R4-Type Extensions

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Slide 1: RV32I R-Type Instruction Format

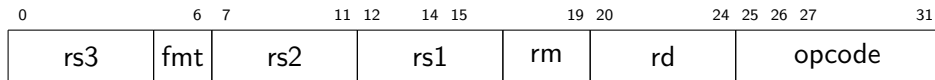
The standard R-Type format is used for register-register arithmetic/logical operations.



- **Opcode (7 bits):** Identifies the instruction class (0110011).
- **rd (5 bits):** Destination register.
- **funct3/7:** Provides 10 bits of additional operation space.
- **rs1/rs2 (5 bits each):** Source register addresses.

Slide 2: R4-Type Instruction Format

Used primarily in the Floating-Point (F/D) extensions for Fused Multiply-Add.



- **rs3 (5 bits):** The third source register (Bits 31–27).
- **fmt/rm:** Replaces funct7/funct3 for FP precision and rounding modes.
- **Three Inputs:** Allows $rd = (rs1 \times rs2) + rs3$.

Slide 3: Bit Allocation Comparison

Feature	R-Type (RV32I)	R4-Type (RV32F)
Source Registers	2 (rs1, rs2)	3 (rs1, rs2, rs3)
Function Bits	10 (funct7 + funct3)	5 (fmt + rm)
Register Width	5 bits (32 regs)	5 bits (32 regs)
Primary Use	Integer ALU	Fused Multiply-Add

Note: R4-type sacrifices function bits (opcode space) to accommodate the address for the third source register.

Slide 4: Why use a Third Input (rs3)?

The *rs3* field enables **Fused Multiply-Add (FMA)** operations:

- **Operation:** $rd = (rs1 \times rs2) + rs3$
- **Throughput:** Replaces two instructions (MUL then ADD) with one.
- **DSP/Graphics:** Essential for dot products and coordinate transformations.

Slide 5: Architectural Impact

Implementing R4-Type instructions requires changes to the Data Path:

- **Register File:** Must be expanded from 2 Read Ports to **3 Read Ports**.
- **Decoder:** Must be able to route three register addresses simultaneously.
- **ALU:** Requires a specialized Fused Multiply-Accumulate (MAC) unit.
- **Wiring:** Significant increase in routing complexity.